

Maithi-Net: A Customized Convolution Approach for Fake News Detection using Maithili Language

Debendra Muduli^{1*}, Santosh Kumar Sharma (Graduate Student Member, IEEE)¹, Dinesh Kumar¹, Akshat Singh¹, Shubham Kumar Srivastav¹

¹*C.V Raman Global University, Bhubaneswar, Odisha*

¹debendra.muduli@cgu-odisha.ac.in

Abstract— Online news consumers now face serious difficulties due to the widespread distribution of fake news on social media platforms. To distinguish fake news from real, the paper suggested a customized CNN model named “Maithi-Net”. The model being suggested is composed of five convolution layers which are capable of automatically acquiring the distinguishing features essential for identifying fake news. Both the CGU-Maithili and ISOT fake news datasets have been used to successfully validate the proposed model. The efficacy of the model is verified with several evolution metrics like accuracy, specificity, sensitivity and F1 score. The model provides the detection accuracy 96.85 % for CGU-Maithili and 97.28% for ISOT fake news datasets. The experimental results show substantial gains over prior state-of-the-art results in the area of fake news detection and validate the potential of our method for categorising misinformation spread via social media.

Keywords— CNN, Fake News, Recurrent Neural Network, Logistic Regression, Naive Bayes

I. INTRODUCTION

Fake news detection has become increasingly important in recent years due to the rise of social media and the ease with which false information can be spread online. Inaccurate information can have serious consequences, including influencing public opinion, causing panic, and undermining the credibility of legitimate news sources. Because of the rise in popularity of social media and other online news sources, the problem of disseminating false information has become increasingly pressing in modern society. Individuals, groups, and even nations can all suffer from the effects of false information. In order to stop the spread of false information and protect the credibility of news organizations, detecting fake news has become an absolute necessity. While there have been several efforts to develop fake news detection models in various languages, there is still a need for more comprehensive and accurate models, particularly in less commonly spoken languages [1,4]. Since then, efforts to create and enhance methods for identifying fake news have increased in popularity. Researchers have used various approaches, including natural language processing, machine learning, and deep learning, to identify and classify fake news. These techniques often rely on features such as the language and style of the text, the credibility of the source, and the level of emotional content in the news article. However, fake news detection remains a challenging problem, as it requires not only identifying false information but also distinguishing it

from genuine news articles. Furthermore, there is a continuous need to develop and refine detection techniques because fake news evolves and adapts as it spreads. In spite of this, it is anticipated that more precise and trustworthy methods for detecting fake news will continue to emerge with the ongoing development of deep learning algorithms [5, 6].

Using machine learning techniques and convolutional neural networks (CNNs), we demonstrate here the creation of a fake news detection model for the Maithili language. Our proposed model aims to identify fake news articles from genuine ones by leveraging supervised learning techniques and deep learning architecture. After being fed a dataset of Maithili articles to train on, the model proved highly effective at identifying instances of disinformation. In addition to proving the efficacy of the proposed model for detecting fake news in less widely spoken languages, the suggested model's performance also illustrates the model's potential for extension to other languages.

II. RELATED WORKS

Tanvir et al.[1] have contributed to machine learning and deep learning-based fake news detecting systems. They have employed different classifiers like SVM (Support Vector Machine), RNN (Recurrent Neural Network), NB (Naive Bayes) and LR (Logistic Regression). Priyadarshi et al.[2] have proposed a maithili POS tragger based system, with condition random field classifier. Granik et al.[3] the paper contributes to this area by suggesting a straightforward method based on the Naive Bayes classifier. Classification accuracy on the test set was close to 74% once the method was turned into a software system and evaluated against a dataset of Facebook news postings. Despite the modesty of this outcome, the research details many methods by which it could be enhanced. This study is significant because it demonstrates that artificial intelligence techniques can be used to tackle the issue of detecting fake news. Kaliyar et al.[4] have addressed this issue by proposing a deep learning approach, called FakeBERT, that employs a bidirectional training approach using a combination of a single-layer deep Convolutional Neural Network (CNN) and the Bidirectional Encoder Representations from Transformers (BERT). To effectively detect fake news, the suggested method attempts to capture semantic and long-distance relationships in phrases. The study expands upon prior work that used sequential neural networks to encode both content and social context-level information, and it emphasizes the significance of

bidirectional training for modelling pertinent information in fake news identification.

In order to classify bogus news, Nasir et al.[5] devised a hybrid deep learning model that blends convolutional and recurrent neural networks. Their model was developed and evaluated on two fake news datasets (ISO and FA-KES) and achieved much better detection results than prior non-hybrid baseline techniques. Further tests were done to verify the suggested model's generalizability across diverse datasets, and the findings were encouraging. Using a combination of convolutional neural networks (CNNs) and long-short term recurrent neural networks, Oluwaseun et al.[6] established a methodology to identify and categorize false news stories shared on Twitter. (LSTMs). With their proposed method, they were able to identify important features related to fake news stories with an accuracy of 82% and without the use of any specific domain knowledge. Iftikhar et al.[7] proposed a model using an algorithmic system for determining whether or not news stories are fraudulently using an ensemble method of machine learning. The authors developed an ensemble of machine learning algorithms based on their findings about the textual features that help distinguish fake from true news.

Hadi et al.[8] proposed using margin loss and various embedding methods in Convolutional Neural Networks (CNNs) for identifying bogus news. The authors compared a fixed word embedding with a trainable one that can adapt to new data. ISOT and LIAR are two recent examples of popular datasets that were used to test the proposed designs. Results were encouraging, with an improvement of 7.9% on ISOT and 2.1% on the LIAR dataset's test set compared to state-of-the-art approaches. In this paper [9], Ray and colleagues provide a comprehensive review of concerns in Natural Language Processing (NLP) connected to the detection of bogus news and related tasks. They comprehensively evaluate existing natural language processing (NLP) methods, datasets, and variations on the original problem design. In light of these findings, they propose next steps for developing more accurate and useful detection models. The authors also stress the contrasts between fake news identification and other similar tasks and the significance of NLP solutions for this

problem. Wang et al.[10] have developed a new, openly available LIAR dataset for identifying hoaxes in the media. The dataset can be used for fact-checking studies, and it contains 12.8K brief statements from POLITIFACT.COM that have been hand annotated over the course of a decade. For the purpose of bettering a deep learning model based solely on text, the authors have developed a novel, hybrid convolutional neural network that incorporates metadata with text. They demonstrated that the hybrid method was successful in identifying fake news by analysing its surface-level linguistic patterns.

Overall, these recent research papers show that machine learning and deep learning techniques can be effective for fake news detection. However, there are still many challenges in this field, such as dealing with evolving and dynamic fake news, dealing with bias in datasets, and explaining the decisions made by these models.

III. PROPOSED SYSTEM

The suggested method for creating a model to identify bogus news in Maithili language involves leveraging machine learning approaches and a customised CNN model named Maithi-Net for classifying the news articles. Figure 1 depicts the proposed model work flow, and the various steps are discussed in more detail below.

A. Architecture of the system

The proposed method takes as input a series of news headlines and labels indicating whether or not each headline is true. After that, they are transformed into a skill record for use throughout the training phase. Pre-processing is the transformation that involves a chain of operations like stemming and vectorization. Training and testing sets are created from the pre-processed dataset. The training module uses the training dataset to generate a decision module that can be applied to the test dataset. The model is accepted it can be retained and utilised and then the training stops.

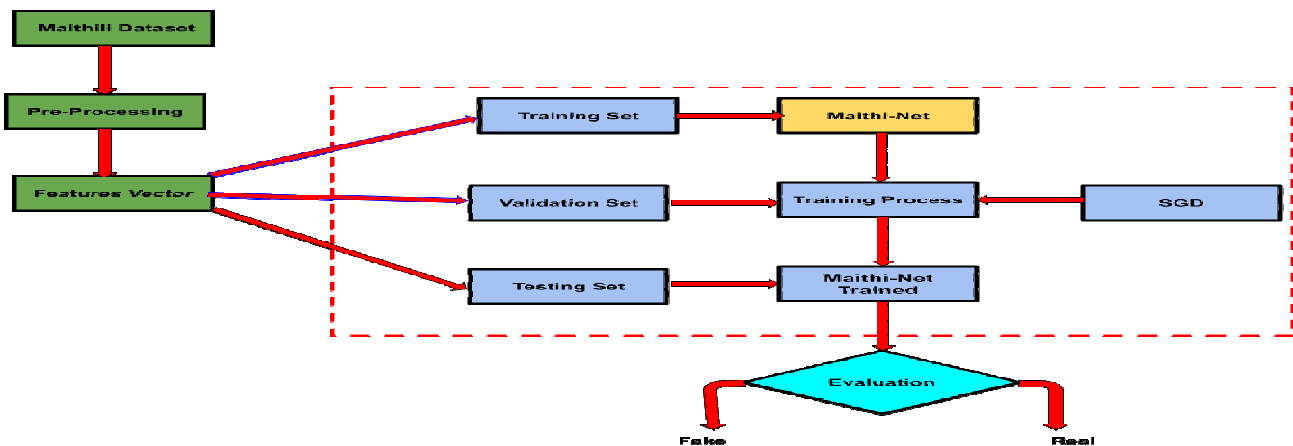


Fig. 1 ARCHITECTURE OF THE FAKE NEWS DETECTION MODEL

B. Pre-processing

If you are working with a dataset that includes news headlines in the Maithili language for your research paper, you would need to perform several pre-processing steps to prepare the data for analysis.

I. Data Cleaning:

Remove irrelevant information like tags, special characters, punctuations, or any other non-alphanumeric characters that are not required for analysis.

II. Text Tokenization:

The Maithili language uses a different script and has different rules for tokenization. You may need to use language-specific tokenizers that can split the news headlines into words, phrases, or sentences.

III. Stopwords Removal:

Stopwords are commonly occurring words which do not add much value to the meaning of the text. Eliminating them can help you analyse data more quickly by lowering its dimensionality.

IV. Stemming/Lemmatization:

Maithili has a complex morphology and many inflections. We may need to use language-specific stemming or lemmatization techniques to normalize the words to their base form. This will help reduce the number of unique words, making the analysis more manageable.

V. Text Encoding:

Once the data has been pre-processed, we have encoded the text into a numerical format. We have applied Word2vec[11], which is a widely used technique for generating word embedding that captures the semantic meaning of words. It involves training a neural network on a large corpus of text data to learn the vector representations of words in a low-dimensional space. These vector representations can be used to capture the meaning of words and their relationships with other words.

VI. Normalization:

The Word2vec produces a set of high-dimensional vector representations for each word in the dataset. These vectors can be used to represent the semantic meaning of the words and can be input into a machine-learning model for further processing. However, the range of values for the vector components can vary widely depending on the specific word and the context it appears. This can make it difficult to compare or combine the vectors in a meaningful way. To address this issue, you can apply a normalization technique such as min-max normalization to scale the vector components to a common range. Min-max normalization scales the values in the vector to the range [0,1], where the minimum value in the vector is mapped to 0, and the maximum value is mapped to 1. This can help to standardize the input data and improve the performance of the machine learning model. The formula for min-max normalization is:

$$A_{\text{Normalized}} = (A - \text{Min}(A)) / (\text{Max}(A) - \text{Min}(A))$$

A normalized is the normalised value, where A is the original value, Min(A) is the dataset's minimum value, and Max(A) is its maximum value.

VII. Data Splitting:

Once the data has been encoded, it can be divided into a training set and a test set. In this work we have considered the training and testing set with the ratio 80% and 20%. After the pre-processing operation, we have used a customized CNN model for automated feature extracting and classification.

C. Convolution Neural Network(CNN)

Detecting fake news is a challenging task that requires a combination of natural language processing and machine learning techniques. Convolutional Neural Networks (CNNs) have been shown to be effective in various natural language processing tasks, including sentiment analysis, text classification, and language translation. During training, the CNN is fed a set of labeled data that consists of both real and fake news articles. The CNN adjusts its weights based on the loss function, which measures the difference between the predicted and actual labels. Once the CNN is trained, it can be used to predict whether a new article is real or fake based on its text content.

I. Convolution:

Convolution is a powerful technique that can be used in fake news detection in Maithili language to help combat the spread of false information and promote accurate reporting. The convolutional layer will apply a set of filters or kernels to the input matrix, sliding over the matrix in a specified stride and performing element-wise multiplication and addition operations between the filter and the local input regions. The resulting output is a feature map that represents the extracted features from the input data [12].

II. Pooling:

Max Pooling involves dividing the feature map into a set of rectangular areas and taking the maximum value of each area. This reduces the dimensionality of the feature map while preserving the most important features. Max Pooling can be useful in fake news detection in Maithili language by highlighting the most important features that are indicative of fake news.

III. Batch Normalization:

Deep learning models, such as Convolutional Neural Networks (CNNs), employ the batch normalisation technique, which can be used to identify bogus news in Maithili. To enhance the performance and stability of the model, it entails normalising the output of each layer.

IV. Activation function:

The activation functions are used to introduce non-linearity into the model, allowing it to learn complex relationships between the input and output data. Here we have used Relu activation function. It is the most widely used activation function in CNNs. It replaces all negative values in the output with zero, which helps to introduce non-linearity and sparsity in the output.

V. Stride:

It allows the CNN to capture more detailed patterns in the text, such as specific phrases or sentence structures that are indicative of fake news.

VI. Fully Connected:

Convolutional Neural Networks (CNNs) utilized for fake news identification in Maithili are examples of deep learning models which incorporate Fully Connected layers, also known as dense layers.

VII. Soft Max:

When training a neural network to detect bogus news in Maithili, the activation function softmax is frequently utilized at the network's output layer. The softmax function takes a vector of inputs and normalizes them to produce a probability distribution over multiple classes.

VIII. Maithi-Net Deep Learned Model:

Specifically, the "Maithi-Net" deep learning model uses four convolution layers and a single fully-connected layer to achieve its goals. The first layer of the proposed model named CON_1 consists of 64 filters with a size of 5x5. Subsequently, batch normalisation, Relu as an activation layer, and the max-pooling layer have been employed. The pooling layer uses a filter of size 2x2 and a stride of 2 to decrease the feature map's dimensionality. Similarly, the second convolution layer named CON_2 contains 64 filters with size 3x3. Subsequently, the third (CON_3), fourth (CON_4) and fifth (CON_5) convolution layers contain 32, 16, and 8 filters with size 3x3, and the rest of the layers same as the first convolution layer. Once all the deep-learned features have been combined into a single vector at the fully connected layer, the resulting prediction of fake or real news can be made at the softmax layer. The detailed architecture of the "Maithi-Net" is tabulated in Table I.

TABLE I
THE SPECIFIC SETTINGS OF THE PROPOSED MAITHI-NET

L_a	S_z	F_s	N_f	Stride
I_N	12000×1			
Con_1	$C_L + B_N + R_{ELU}$	5x5	64	1x1
M_{PL_1}		2x2		2x2
Con_2	$C_L + B_N + R_{ELU}$	3x3	64	1x1
M_{PL_2}		2x2		2x2
Con_3	$C_L + B_N + R_{ELU}$	3x3	32	1x1
M_{PL_3}		2x2		2x2
Con_4	$C_L + B_N + R_{ELU}$	3x3	16	1x1
M_{PL_4}		2x2		2x2
Con_5	$C_L + B_N + R_{ELU}$	3x3	8	1x1
Fully Connected	OP Size =2			
OP	Classification Layer	Soft-Max		

* L_a -Layer, S_z - size, I_N -Input, F_s -Filter Size, N_f - No of Filter, C_L -Convolution, B_N -Batch Normalization, R_{ELU} -ReLU, M_{PL} -Max-pooling layers, OP-Output

IV. EXPERIMENT AND RESULTS

a. Used Dataset

The proposed deep-learned CNN model is evaluated with two datasets named as CGU_Maithili and ISOT fake news dataset. We have created the CGU_Maithili dataset collected from various web resources. It consists of a collection of news

articles written in Maithili. The articles have been collected from various Maithili e-news websites like "Maithilijindabaad", and "Maithili Dainik", and e-journals like "Baidehi" and "Sahitya Academy". The dataset consists of 20,000 instances, with 10,000 number of real and 10,000 number of fake news. The ISOT Fake News Dataset has been collected from Kaggle (<https://www.kaggle.com/datasets/csmalarkodi/isot-fake-news-dataset>). Here, we have considered 40,000 number of instances with 20,000 fake and 20,000 number of real news.

b. Result And Discussion

Experimental simulations were run on a device equipped with an Intel Xeon 2.20 GHz processor, 16 GB of RAM, and an NVIDIA 11 GB GPU to verify the correctness of the suggested model. For this suggested model, we have been using Python-3.11.0. First, we have loaded the dataset into a Pandas Data Frame. After that, pre-process the text data by removing stop words, punctuation, and special characters. Convert all text to lowercase and perform stemming or lemmatization to normalize the words. Then we used word2vec to convert the text into word embedding. We have applied min-max normalization techniques to scale the vector components to a common range of 0 to 1. Now, Split the data into training, validation and testing sets with ratio 80:20:20. The division of the datasets have been shown in Table II. The training set is used to train the proposed customized CNN model called "Maithi-Net", while the testing set is used to evaluate its performance. We have fine-tuned the model by adjusting its hyper parameters such as the learning rate, batch size, and the number of epochs. Then, the performance of the model has been tested with a testing set. We have used metrics such as accuracy, precision, recall, and F1-score to measure the model's performance.

In this research work, we have simulated two experiments on two datasets, "CGU_Maithili" and ISOT fake news dataset. The accuracy and loss function are used to evaluate the proposed model's efficacy. We've used a stochastic gradient descent (SGD) optimizer with a 0.0001 learning rate to fine-tune the weights. The categorical cross-entropy loss function has been used to select the optimal weights. The hyperparameters of the proposed model is tabulated in the Table III. In the first experiment, we applied several conventional classifiers like logistic regression, decision tree, K-nearest neighbour, naïve Bayes and support vector machine. In the second experiment, we employed the customized CNN model called "Maithi-Net". The experimental results of the "CGU_Maithili" and ISOT fake news datasets have been tabulated in Table IV and Table V, and the comparison graph has been shown in Figure 8. The accuracy and loss performance graphs are shown in Figures 2-5 for both datasets. Similarly, the confusion matrices of the experiment have been shown in Figures 6 and 7. Also, the experimental results of the proposed model have been compared with the existing models and the results have been shown in Table VI. From the experimental results, we have observed that the proposed "Maithi-Net" is an automated fake news detection

model which provides better classification results as compared to other state-of-art models with less number of tunable parameters.

TABLE II
THE TRAINING AND TESTING SPLITTING OF DATASETS

Datasets	Training set		Validation Set		Testing set	
	Real	Fake	Real	Fake	Real	Fake
CGU_Maithili	6000	6000	2000	2000	2000	2000
ISOT fake news	12000	12000	4000	4000	4000	4000

TABLE III
HYPER-PARAMETERS OF MAITHI-NET MODEL

Hyper parameters	Value
Batch Size	120
Learning Rate	0.0001
Epochs	100

TABLE IV
DIFFERENT CLASSIFIERS RESULTS ON THE CGU_MAITHILI

Classifier	Accuracy	Sensitivity	Specificity	F1_Score
Logistic Regression	72.45	72.40	72.50	72.44
Random Forest	92.33	90.65	94.00	92.19
KNN	93.35	92.25	93.75	93.52
Decision Tree	92.58	90.90	94.25	92.19
AdaBoost	89.00	88.00	90.00	88.89
Maithi-Net (Proposed Model)	96.85	92.20	97.50	96.83

TABLE V
DIFFERENT CLASSIFIERS RESULTS ON THE ISOT FAKE NEWS

Classifier	Accuracy	Sensitivity	Specificity	F1_Score
Logistic Regression	76.88	71.25	82.50	75.50
Random Forest	91.63	95.00	88.25	91.90
KNN	92.48	97.45	87.50	92.83
Decision Tree	92.38	97.00	87.75	92.71
AdaBoost	91.45	95.10	87.80	91.75
Maithi-Net (Proposed Model)	97.28	98.55	96.00	97.31

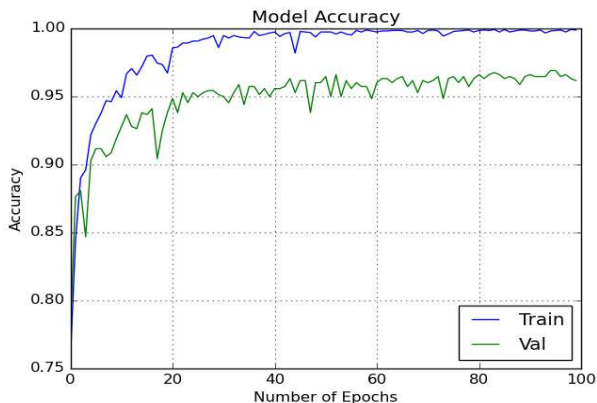


Fig.2. ACCURACY PLOT FOR CGU_MAITHILI DATASET

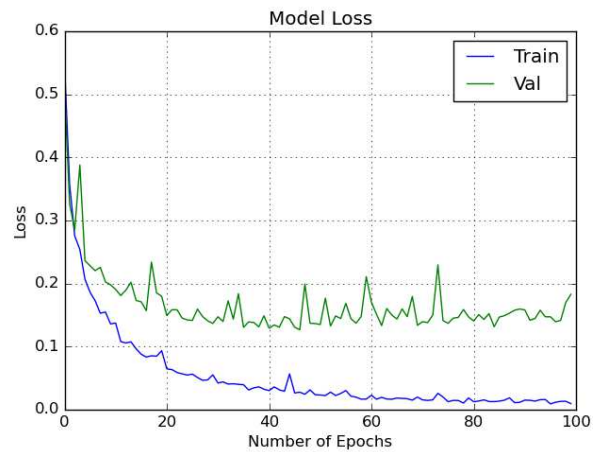


Fig.3. LOSS PLOT FOR CGU_MAITHILI DATASET

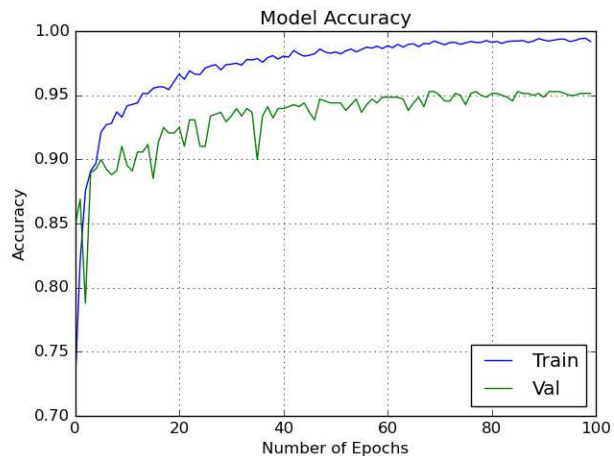


Fig. 4. ACCURACY PLOT FOR ISOT FAKE NEWS DATASET

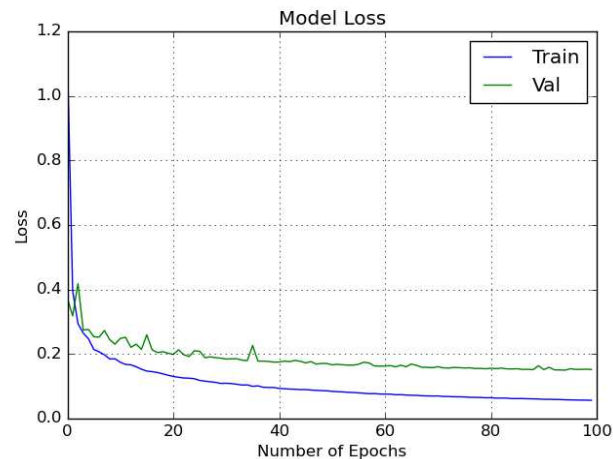


Fig. 5. LOSS PLOT FOR ISOT FAKE NEWS DATASET

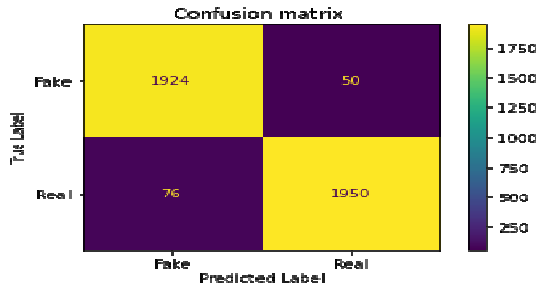


Fig.6. CONFUSION MATRIX FOR CGU_Maithili

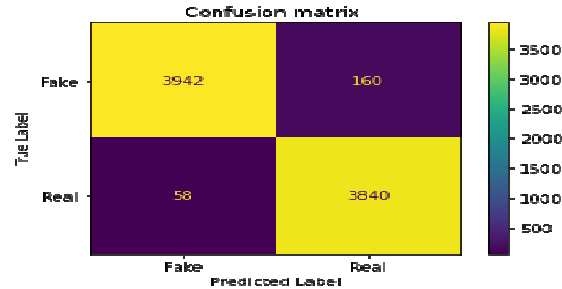


Fig.7. CONFUSION MATRIX FOR CGU_Maithili

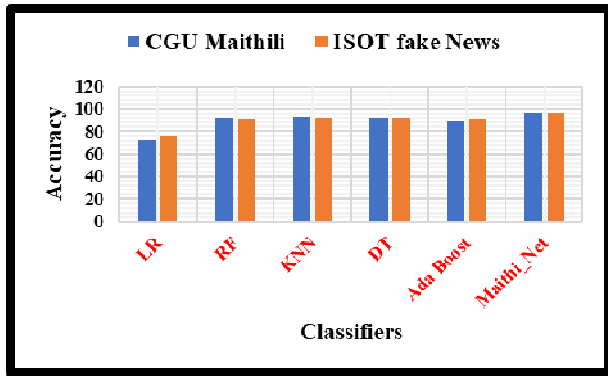


Fig.8. COMPARING DIFFERENT CLASSIFIERS WITH PROPOSED MODEL

TABLE VI
EVALUATE THE MAITHI-NET PERFORMANCE USING CGU-MAITHI DATASET AND COMPARE IT TO OTHER MODELS.

Existing Models	Accuracy (%)	
	ISOT Fake News	CGU-Maithi
Naïve Bayes [1]	89.06	-
SVM [1]	89.34	-
Logistic Regression [1]	69.47	-
RNN [1]	74.00	-
LSTM [1]	78.00	-
CRF [2]	82.67	-
Naïve Bayes [3]	74.00	-
BERT [4]	98.90	-
Hybrid CNN-RNN [5]	60.00	-
Decision Tree [7]	94.00	-
Hybrid CNN [10]	27.00	-
Maithi-Net (Proposed Model)	97.28	96.85

V. CONCLUSIONS

To determine whether or not news reports are authentic, this study proposes an automated, customized deep-learned model called "Maithi-Net.". The proposed model for detecting fake news has five learnable layers and fewer tuning parameters. The deep-learned model is used to capture the high-level features from the input data, such as the specific words or phrases that are commonly used in fake news articles. In addition, several techniques such as batch normalization, pooling, and activation functions can help to improve the performance of the model. The fully connected layers and softmax Functions can also be used to predict the probability of an article being fake or real. The proposed method has been shown to be more effective at identifying fake news than other recently proposed competent methods. We plan to evaluate the model's accuracy using additional Indian language datasets and a variety of word embedding methods in the future.

REFERENCES

- [1] Abdullah-All-Tanvir, Mahir, E. M., Akhter S., & Huq, M. R. (2019). Detecting Fake News using Machine Learning and Deep Learning Algorithms. 7th International Conference on Smart Computing & Communications (ICSCC), Sarawak, Malaysia, Malaysia, 2019, pp.1-5, <https://doi.org/10.1109/ICSCC.2019.8843612>
- [2] Priyadarshi, Ankur, and Sujana Kumar Saha. "Towards the first Maithili part of speech tagger: Resource creation and system development." Computer Speech & Language 62 (2020): 101054.
- [3] Mykhailo Granik and Volodymyr Mesyura. Fake news detection using naive bayes classifier. In 2017 IEEE First Ukraine Conference on Electrical and Computer Engineering (UKRCON), pages 900–903. IEEE, 2017.
- [4] Kaliyar, Rohit Kumar, Anurag Goswami, and Pratik Narang. "FakeBERT: Fake news detection in social media with a BERT-based deep learning approach." Multimedia tools and applications 80.8 (2021): 11765-11788.
- [5] Nasir, Jamal Abdul, Osama Subhani Khan, and Iraklis Varlamis. "Fake news detection: A hybrid CNN-RNN based deep learning approach." International Journal of Information Management Data Insights 1.1 (2021): 100007.
- [6] Ajao, Oluwaseun, Deepayan Bhowmik, and Shahrzad Zargari. "Fake news identification on twitter with hybrid cnn and rnn models." Proceedings of the 9th international conference on social media and society. 2018.
- [7] Ahmad, Ifikhar, et al. "Fake news detection using machine learning ensemble methods." Complexity 2020 (2020): 1-11.
- [8] Goldani, Mohammad Hadi, Reza Safabakhsh, and Saeedeh Momtazi. "Convolutional neural network with margin loss for fake news detection." Information Processing & Management 58.1 (2021): 102418.
- [9] Oshikawa, Ray, Jing Qian, and William Yang Wang. "A survey on natural language processing for fake news detection." arXiv preprint arXiv:1811.00770 (2018).
- [10] Wang, William Yang. "liar, liar pants on fire": A new benchmark dataset for fake news detection." arXiv preprint arXiv:1705.00648 (2017).
- [11] Goldberg, Yoav, and Omer Levy. "word2vec Explained: deriving Mikolov et al.'s negative-sampling word-embedding method." arXiv preprint arXiv:1402.3722 (2014).
- [12] Muduli, Debendra, Ratnakar Dash, and Banshidhar Majhi. "Automated diagnosis of breast cancer using multi-modal datasets: A deep convolution neural network-based approach." Biomedical Signal Processing and Control 71 (2022): 102825.